Sharik Anwar Zahir Hussain *Systems & AI Engineer*

+49 17645971872 zahirhussain anwar-prog

zsharikanwar@gmail.com zahir-portfolio.dev

SUMMARY

Systems and AI engineer with a Master's degree in Artificial Intelligence from Technische Hochschule Würzburg-Schweinfurt. Experienced in building and supporting real-world AI systems, backend services, and automation workflows. Skilled in debugging, analyzing issues, and ensuring reliable execution in production environments. Strong foundation in Python, C++, and API integration. Fluent in English and advancing German proficiency toward B2 level.

PROFESSIONAL EXPERIENCE

Reborse Startup Jun 2025 – Present | Würzburg

AI Engineer

- Developed backend services and automation workflows, ensuring stable system behavior and improving operational efficiency in production environments.
- Built and integrated AI chatbot solutions, handling real-time data processing and resolving integration issues to maintain consistent system performance.
- Tested, monitored, and improved AI-based features in practical environments, focusing on debugging, reliability, and consistent execution.

Technische Hochschule Würzburg Schweinfurt Dec 2025 – Apr 2026 | Würzburg

Research Assistant – AI Systems Integration (Part-time)

- Developed and integrated AI-based automation workflows within web applications, focusing on system interaction, issue analysis, and stable execution.
- Built REST APIs and backend processes for calendar systems using Node.js, enabling structured communication and reliable system behavior.
- Supported deployment and integration tasks, including HTTPS configuration and system setup, ensuring stable and production-ready workflows.

SKILLS

Programming - Proficient
Python, C++, SQL, NumPy, Pandas

Automation & Systems - Competent
REST APIs, Backend Development, Git, Docker, API Integration

Actively Learning
Linux systems (CLI, debugging basics)

AI & Machine Learning - Proficient
LLMs, AI Workflows, Neural Networks, Predictive Modeling

LANGUAGES

German
B2 (in progress)

English
C2

Tamil
Native

PROJECTS

BewerbAI : AI-Powered Job Application Assistant Jan 2026 – Mar 2026

- Built a full-stack application using AI tools to automate job description analysis and generate tailored CVs and cover letters.
- Implemented PDF parsing, bilingual workflows, and modular automation pipelines for structured data processing and system interaction.
- Deployed the system with secure backend API handling, ensuring reliable execution and consistent system performance.

AI-Based Heat Demand Forecasting (Master's Thesis) Apr 2025 – Oct 2025

- Developed machine learning models for prediction tasks using structured datasets and evaluated their performance through systematic analysis.
- Built data processing pipelines to ensure consistent data handling and reliable model evaluation across experiments.
- Explored reinforcement learning methods for optimization in automated decision-making scenarios.

Reinforcement Learning Approaches to Blackjack Apr 2024 – Jul 2024

- Built a Q-learning agent to model decision-making in a stochastic blackjack environment.
- Integrated card counting methods (Hi-Lo, Zen) and rule variations to capture different gameplay dynamics.
- Conducted simulation-based experiments to evaluate policy performance and benchmark against standard blackjack strategies.

3D Object Detection in Manufacturing Environments May 2023 – Feb 2024

- Developed a CNN-based inspection system for classifying manufacturing components using real and synthetic image data.
- Combined industrial images with synthetic 3D-CAD data and applied transfer learning (VGG ImageNet) to improve generalization.
- Enhanced model robustness using data augmentation techniques to handle variations in lighting and viewpoints.

PUBLICATIONS

Automated Sugarcane Disease Detection Using Faster RCNN with an Android Application Apr 2022

IEEE

Designed and implemented a Faster R-CNN–based object detection system for sugarcane disease detection using ~1,500 labeled images, and deployed it as an Android application enabling real-time inference with reliable performance on field data.

EDUCATION

Master's in Artificial Intelligence Mar 2023 – Jan 2026 | Würzburg

Technical University of Applied Sciences Würzburg-Schweinfurt (THWS)

Bachelor's in Computer Science and Engineering May 2018 – Apr 2022 | India

Kalasalingam University